QUASI-SPLITTING EXISTS AFTER AN ETALE LOCALIZATION

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Editorial note. No omission markers within this span. Dedup: parallel to the strong and plain siblings; conclusion differs (quasi-split), counted separately.

Allowed standard prerequisites (editorial).

- The universal finite part of a groupoid. lemma-finite-part-groupoid (same chapter, NOT carded, declared external). Canonical construction of U', Z_univ, and the point u'.
- Splitting siblings. lemma-strong-splitting (carded L26658), lemma-splitting (carded L26659). Same architecture; here the support shrinks to the unit e(u).

Source reference. Original lemma 04RU: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2356–2356. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 04RT: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2256–2256. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04RV: sources/stacks-spaces-more-groupoids/pinned-original-source.tex, source lines 2493–2493. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1 (Existence of quasi-splitting)**.** *In Situation 04RV there exists an algebraic space U', an étale morphism $U' \to U$, and a point $u' : \mathrm{Spec}(\kappa(u)) \to U'$ lying over $u : \mathrm{Spec}(\kappa(u)) \to U$ such that the restriction $R' = R|_{U'}$ of R to U' is quasi-split over u'.*

Proof. Let $f : (U', Z_{univ}, s', t', c') \to (U, R, s, t, c)$ be as constructed in Lemma 04RU. Recall that $R' = R \times_{(U \times_S U)} (U' \times_S U')$. Thus we get a morphism $(f, t', s') : Z_{univ} \to R'$ of groupoids in algebraic spaces

$$(U', Z_{univ}, s', t', c') \to (U', R', s', t', c')$$

(by abuse of notation we indicate the morphisms in the two groupoids by the same symbols). Now, as $Z_{univ} \subset R \times_{s,U,g} U'$ is open and $R' \to R \times_{s,U,g} U'$ is étale

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(as a base change of $U' \to U$) we see that $Z_{univ} \to R'$ is an open immersion. By construction the morphisms $s', t' : Z_{univ} \to U'$ are finite. It remains to find the point u' of U'.

We think of u as a morphism $\mathrm{Spec}(\kappa(u)) \to U$ as in the statement of the lemma. Set $F_u = R \times_{s,U} \mathrm{Spec}(\kappa(u))$. The morphism $F_u \to \mathrm{Spec}(\kappa(u))$ is quasi-finite at $e(u)$ by assumption. Hence we can find a decomposition into open and closed subschemes

$$F_u = Z_u \amalg Rest$$

for some scheme Z_u finite over $\kappa(u)$ whose support is $e(u)$. Hence by the construction of U' in Section 04RT (u, Z_u) defines a $\mathrm{Spec}(\kappa(u))$-valued point u' of U'. To finish the proof we have to show that $e'(u') \in Z_{univ}$ which is clear. $\square$